DOCUMENT C | SOLUTIONS 1

Worked solutions and proposed marking scheme

Best-fit predicted practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 1 | Question 1: Image quality and nuclear medicine

1(a) [3 marks]

Question. Define spatial resolution. State what the point spread function (PSF) represents and write the image-formation relationship for a linear, shift-invariant system.

Expected answer and working. Spatial resolution is the minimum separation at which distinct small features can be resolved. The PSF h is the image of an ideal point object. For a linear, shift-invariant system, I(x,y)=O(x,y)*h(x,y), with * denoting convolution. A finite-width PSF spreads object signal over neighbouring positions.

Proposed allocation: Resolution definition (1); Point response (1); Convolution with stated assumption (1). **Total 3.**

Source map: S1 pp.39,45-48. **Closest past question:** 2025 Q1(a)i-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. A line spread function is L(x)=C exp[-(x-x0)^2/(2σ^2)]. Derive its full width at half maximum (FWHM). Evaluate the width for σ=0.60 mm.

Expected answer and working. At the peak L(x0)=C. At each half-height point, exp[-(x-x0)^2/(2σ²)]=1/2. Taking natural logs gives (x-x0)²=2σ² ln2, so the two points are x0±σ√(2 ln2). Their separation is FWHM=2√(2 ln2)σ=2.35482σ. For σ=0.60 mm, FWHM=1.41289 mm, approximately 1.41 mm.

Proposed allocation: Half-height equation (1); Logarithm and two roots (1); Full-width expression (1); 1.41 mm with unit (1). **Total 4.**

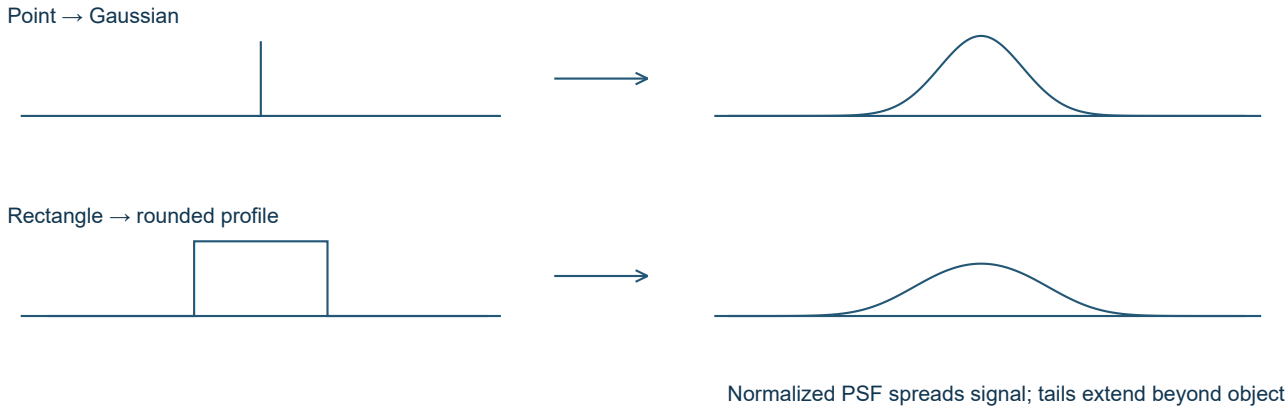
Source map: S1 pp.41-44. **Closest past question:** 2024 Q1(a)iii. **Status:** Recurring family.

Typical mark-losing error: Do not report the half-width σ√(2 ln2) as FWHM.

1(c) [3 marks]

Question. Sketch the result of convolving (i) an ideal point and (ii) a narrow rectangular object with a normalized Gaussian PSF. Explain the effect on fine structures.

Expected answer and working. An ideal point becomes the Gaussian PSF (scaled by its point strength). The rectangle has rounded edges, nonzero tails and a reduced peak when it is narrow relative to the PSF. Convolution preserves integrated signal for a unit-area PSF, but spreads it; closely spaced fine features can merge.



Proposed allocation: Point becomes Gaussian (1); Rounded rectangle and tails (1); Loss of fine detail / merging (1). **Total 3.**

Source map: S1 pp.45-48. **Closest past question:** 2025 Q1(a)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [4 marks]

Question. State four desirable properties of a SPECT radiotracer, explaining the imaging reason for each.

Expected answer and working. Choose a physical half-life long enough for distribution and acquisition but short enough to limit unnecessary exposure; predominantly monoenergetic gamma emission with little unnecessary particulate dose; gamma energy suitable for tissue escape and collimator/detector performance (the lecture suggests roughly 100-200 keV); and high target uptake with low background uptake to improve contrast.

Proposed allocation: Suitable half-life + reason (1); Gamma emission + reason (1); Suitable energy + reason (1); Target/background specificity + reason (1). **Total 4.**

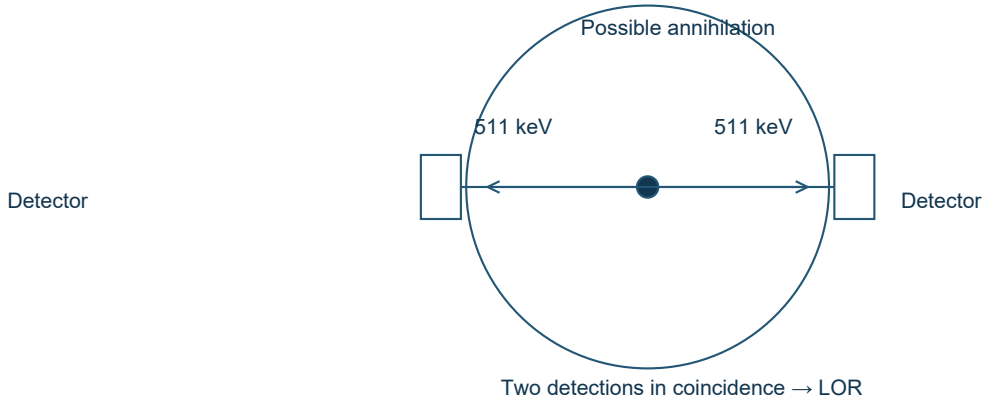
Source map: S06 pp.9-10. **Closest past question:** 2024 Q1(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. Using a labeled sketch, explain positron annihilation coincidence detection in PET. Give two benefits of combining PET with CT.

Expected answer and working. The tracer emits a positron which slows in tissue and annihilates with an electron. The dominant two-photon event produces two approximately opposite 511 keV photons. Opposing detector signals accepted within a coincidence time window define a line of response (LOR), rather than an exact point. Many LORs reconstruct tracer distribution. CT supplies anatomical localization and an attenuation map for PET correction.



Proposed allocation: Positron annihilation and 511 keV pair (1); Opposing detectors + coincidence window (1); LOR sketch / reconstruction (1); Anatomical localization (1); Attenuation correction (1). **Total 5.**

Source map: S06 pp.25-34. **Closest past question:** 2025 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [6 marks]

Question. At t=0, two patients have in-body activities AA=240 MBq and AB=80 MBq. A has physical and biological half-lives of 6 h and 6 h. B has physical and biological half-lives of 12 h and 12 h. Assuming independent exponential clearance, find when their in-body activities are equal and the common activity.

Expected answer and working. Independent physical decay and clearance add their rate constants: 1/Teff=1/Tphys+1/Tbio. Thus Teff,A=3 h and Teff,B=6 h. Set 240·2^(-t/3)=80·2^(-t/6). Dividing gives 3=2^(t/6), hence t=6 ln3/ln2=9.50978 h. Substitution into B gives AB=80/3=26.6667 MBq. A starts higher but decreases faster; the equality is a single positive crossing.

Proposed allocation: Effective-rate law (1); Both effective half-lives (1); Equality equation with initial ratio (1); Log solution (1); 9.51 h (1); 26.7 MBq and interpretation (1). **Total 6.**

Source map: S06 pp.6-8. **Closest past question:** 2024 Q1(b)iv. **Status:** Recurring family.

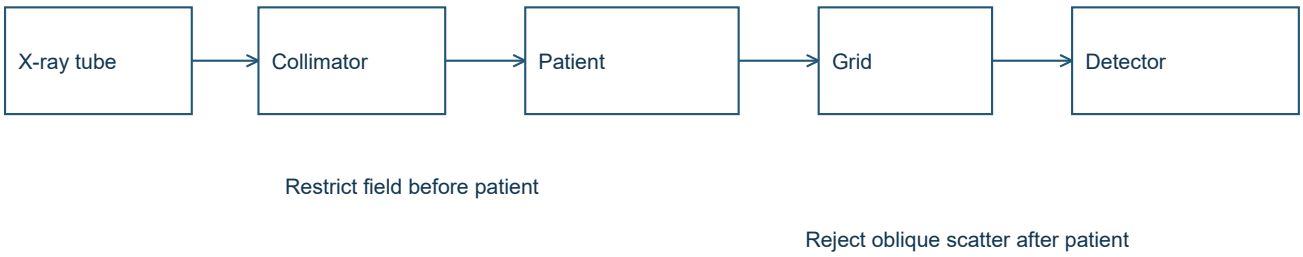
Typical mark-losing error: Use effective half-lives for in-body activity; do not add the half-lives themselves.

Model 1 | Question 2: Projection radiography and computed tomography

2(a) [4 marks]

Question. Draw a planar X-ray radiography system, labeling the tube, beam collimator, patient, anti-scatter grid and detector. State the distinct functions of collimator and grid.

Expected answer and working. The tube produces X-rays; the upstream collimator restricts the irradiated field before the patient. The downstream grid preferentially absorbs obliquely scattered photons before they reach the detector. The detector converts transmitted radiation to a recordable image. Collimation reduces unnecessary exposure and scatter generation; the grid reduces detected scatter but also absorbs some useful primary photons.



Proposed allocation: Correct source-to-detector order (1); Tube and detector functions (1); Collimator restricts field (1); Grid rejects scatter (1). **Total 4.**

Source map: S2 pp.32-42. **Closest past question:** 2025 Q2(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [3 marks]

Question. State three consequences or trade-offs of reducing the effective focal spot. Include an image-quality consequence.

Expected answer and working. A smaller effective spot reduces geometric penumbra and improves spatial resolution. Achieving it by decreasing the anode angle reduces useful beam coverage. Achieving it by concentrating electron power into a smaller actual target area constrains tube loading because of local heating. Credit comparable physically justified consequences; distinguish actual and projected spot size.

Proposed allocation: Less geometric blur / improved detail (1); Coverage trade-off with anode angle (1); Heat or permissible loading trade-off (1). **Total 3.**

Source map: S2 pp.7-13,50. **Closest past question:** 2025 Q2(a)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [5 marks]

Question. A monoenergetic beam illuminates 0.60 cm thick bars with HVL=0.20 cm. Gaps have negligible attenuation. Calculate μ, Ibar/Igap, and Weber contrast C=(Igap-Ibar)/Igap. State one reason this simple model is imperfect for a clinical polychromatic beam.

Expected answer and working. HVL=ln2/μ gives μ=0.693147/0.20=3.46574 cm^-1. There are three HVLs in 0.60 cm, so Ibar/Igap=exp(-μ·0.60)=2^-3=0.125. The specified Weber contrast is 1-0.125=0.875 (87.5%). A clinical beam has a spectrum: lower energies are preferentially removed, causing beam hardening and a thickness-dependent effective attenuation; scatter is another valid limitation.

Proposed allocation: HVL relationship (1); μ=3.47 cm^-1 (1); Transmission 0.125 (1); Contrast 0.875 (1); Physical limitation (1). **Total 5.**

Source map: S2 pp.27-31; S3 p.59. **Closest past question:** 2024 Q2(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [3 marks]

Question. Give three advantages of CT over planar projection radiography.

Expected answer and working. CT separates anatomy into cross-sectional slices, reducing superposition; it has better ability to distinguish small attenuation differences in soft tissue; and calibrated CT numbers permit quantitative attenuation comparisons and geometrical measurements. These advantages coexist with higher typical radiation exposure and reconstruction requirements.

Proposed allocation: Reduced superposition (1); Improved low-contrast discrimination (1); Quantitative HU or accurate 3D localization (1). **Total 3.**

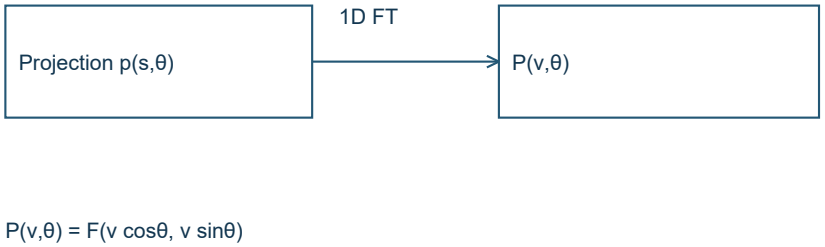
Source map: S3 pp.2-3,31-32. **Closest past question:** 2024 Q2(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

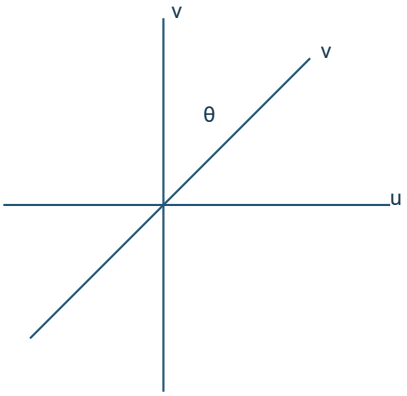
2(e) [4 marks]

Question. State the central section theorem, with a labeled diagram relating projection space to the object Fourier domain.

Expected answer and working. For $p(s,\theta)=\int f(s\cos\theta-t\sin\theta, s\sin\theta+t\cos\theta)dt$, take the 1D FT in s. It equals the 2D FT $F(u,v)$ of the object sampled on the line $(u,v)=(v\cos\theta, v\sin\theta)$ through the origin. The same projection-normal angle θ fixes the radial line. Thus projections at many angles fill radial Fourier samples, followed by interpolation and inverse transformation.



$P(v,\theta) = F(v\cos\theta, v\sin\theta)$



Radial slice of F(u,v)

Proposed allocation: 1D FT of projection (1); Equals radial slice of 2D object FT (1); Same angle / through origin (1); Correct labeled diagram (1). **Total 4.**

Source map: S3 pp.15-16. **Closest past question:** 2024 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(f) [6 marks]

Question. A 2×2 image has unit-weight ray sums: top row 6, bottom row 10, left column 7 and right column 9. Start all pixels at zero. Perform one ART sweep in that order, adding each ray residual equally to its two pixels. Report every intermediate image and determine whether these data uniquely identify all four pixels.

Expected answer and working. Let the image be [[a,b],[c,d]]. After top row: [[3,3],[0,0]]. After bottom row: [[3,3],[5,5]]. The predicted left sum is 8; residual 7-8=-1, so add -0.5 to its two pixels: [[2.5,3],[4.5,5]]. The right sum is 8; residual 9-8=1, so add +0.5: [[2.5,3.5],[4.5,5.5]]. All four sums now match. They do not imply uniqueness: [[a,6-a],[7-a,3+a]] has the same sums. More independent views or constraints are needed to determine the actual image.

Proposed allocation: Two row updates (2); Left residual/update (1); Right residual/final image (1); Verification of sums (1); Nonuniqueness with reason or family (1). **Total 6.**

Source map: S3 pp.26-28. **Closest past question:** 2024 Q2(b)iii. **Status:** Recurring family.

Typical mark-losing error: A fit to a few projections is not proof of a unique reconstruction.

Model 1 | Question 3: Magnetic resonance imaging

3(a) [3 marks]

Question. Name three intrinsic tissue parameters that affect MRI pixel intensity and explain why intensity cannot be interpreted as proton density alone.

Expected answer and working. Proton density ρ, longitudinal relaxation time T1 and transverse relaxation time T2 (or T2* in GE) affect signal. The acquisition TR, TE and flip angle weight their contributions. Two tissues with equal proton density can therefore have different brightness because they recover or dephase at different rates.

Proposed allocation: ρ (1); T1 (1); T2/T2* with sequence dependence (1). **Total 3.**

Source map: S4 pp.3,75,78-79. **Closest past question:** 2024 Q3(a)i; 2025 Q3(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [4 marks]

Question. A 1.5 T scanner uses Gz=20 mT/m to select a 4.0 mm thick axial slice centred at z=0. Calculate the central Larmor frequency, full RF excitation bandwidth and frequency endpoints in Hz.

Expected answer and working. Use γbar=γ/(2π)=42.5771304 MHz/T. Then f0=γbar·1.5=63.8656956 MHz. Convert Gz=0.020 T/m and Δz=0.0040 m. Full Δf=γbar·Gz·Δz=3406.1704 Hz. Endpoints are f0±Δf/2: 63,863,992.48 Hz and 63,867,398.65 Hz (approximately 63.86399-63.86740 MHz). The bandwidth spans the full slab, not just one side.

Proposed allocation: Hz conversion / f0 (1); Correct G and thickness units (1); Full BW=3406 Hz (1); Endpoints f0±1703 Hz (1). **Total 4.**

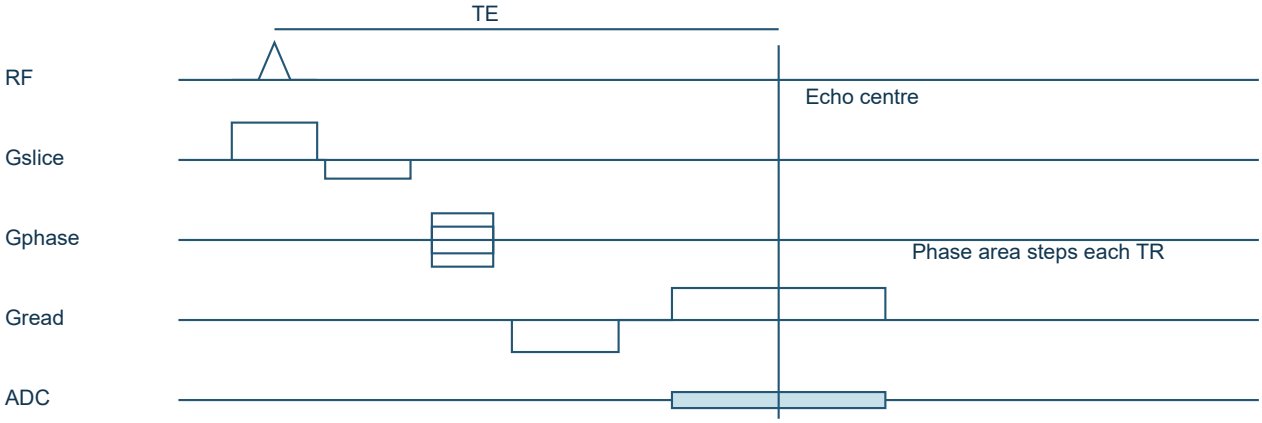
Source map: S4 pp.15,48-54. **Closest past question:** 2023 Q3(b)ii-iii; 2025 Q3(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [5 marks]

Question. Sketch a basic gradient-echo imaging sequence on RF, Gslice, Gphase, Gread and ADC lines. Identify TE and explain the purpose of each gradient.

Expected answer and working. Apply frequency-selective RF concurrently with a slice gradient, then a slice-rephasing lobe. Apply a phase-encoding lobe whose area changes between TR repetitions to select ky. A readout prephaser dephases spins before the opposite-polarity readout gradient produces the echo; acquire data under the readout gradient around its echo centre. TE is RF excitation centre to echo centre. No 180-degree RF refocusing pulse is used in basic GE.



Proposed allocation: RF simultaneous with slice selection (1); Slice rephaser (1); Variable phase-encoding lobe (1); Read prephaser/readout and ADC (1); TE correctly labeled and gradient purposes (1). **Total 5.**

Source map: S4 pp.52-65. **Closest past question:** 2024 Q3(b)i; 2025 Q3(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [4 marks]

Question. For one average, 192 phase-encoding steps and TR=0.80 s, calculate acquisition time. Each slice occupies a complete 20 ms sequence slot. Ignoring overhead and cross-talk, find how many slices can be interleaved without increasing that time. Explain how the selected slice is changed.

Expected answer and working. One phase-encoding line is acquired per TR for each slice. Tscan=Nphase·TR·NEX=192·0.80·1=153.6 s (2 min 33.6 s). A complete 20 ms slot gives floor(800/20)=40 slice slots per TR. Change RF centre frequency under the same slice-selection gradient to excite a different z position; repeat the same phase step for each slice and store separately. The stated full slot, not merely TE, sets this ideal packing bound.

Proposed allocation: 153.6 s (1); 40 slice slots (1); RF centre frequency changes slice position (1); Each slice retains 0.80 s TR and separate data (1). **Total 4.**

Source map: S4 pp.61-74. **Closest past question:** 2025 Q3(b)iii-iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. A spoiled steady-state GE scan has TR=0.50 T1. Find the Ernst angle. For fixed TR, TE and receiver noise, calculate the percentage SNR reduction when a 90° pulse replaces the optimum angle.

Expected answer and working. E1=exp(-TR/T1)=exp(-0.5)=0.60653066. The optimum is αE=acos(E1)=52.66093°. The common ρ·exp(-TE/T2*) factor cancels. At 90°, S90=1-E1=0.39346934. At αE, SE=(1-E1)√(1-E1²)/(1-E1²)=√[(1-E1)/(1+E1)]=0.49489258. Thus S90/SE=√(1-E1²)=0.79506010. With the same noise, SNR loss=100(1-0.79506010)=20.49399%, approximately 20.5%.

Proposed allocation: E1 and optimum angle (1); Correct two signals / ratio setup (1); Ratio 0.7951 (1); Loss 20.5% (1). **Total 4.**

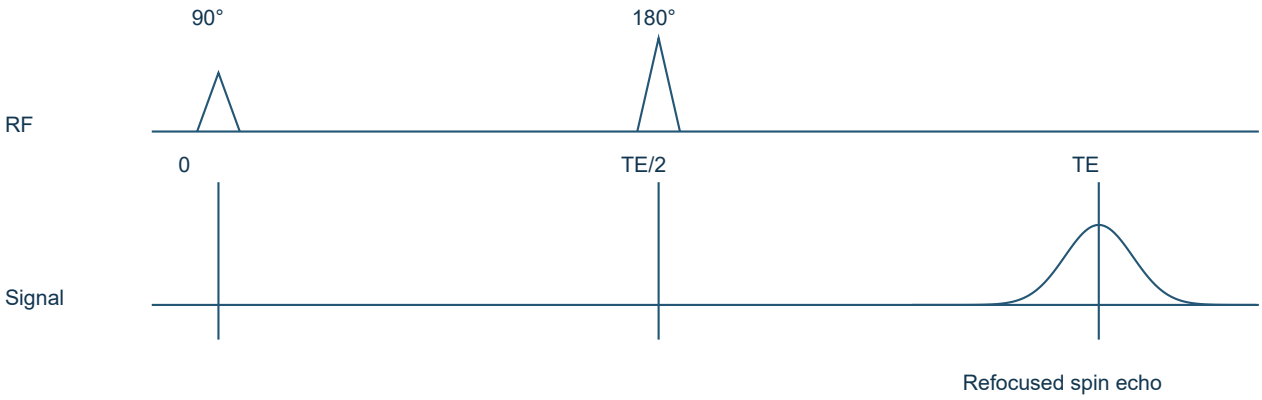
Source map: S4 p.78. **Closest past question:** 2024 Q3(b)iv-v. **Status:** Recurring family.

Typical mark-losing error: Compare signals at identical TR/TE/noise; sin(α) alone omits steady-state saturation.

3(f) [5 marks]

Question. Explain why a conventional spin echo is less sensitive than GE to static magnetic-field inhomogeneity. Sketch the RF and signal timing and state which transverse relaxation time governs the ideal echo envelope.

Expected answer and working. A 90° excitation creates transverse magnetization; static field offsets make spins accumulate different phases. A 180° RF pulse at TE/2 reverses their relative phase positions. Stationary spins with constant offsets rephase at TE, producing an echo. Intrinsic irreversible T2 decay is not recovered. The ideal spin-echo envelope is exp(-TE/T2); gradient echo alone does not reverse static field-offset dephasing and retains T2* effects.



Proposed allocation: Static offsets cause dephasing (1); 180° at TE/2 (1); Rephasing at TE (1); Irreversible T2 remains / SE envelope (1); Labeled RF and signal sketch / GE comparison (1). **Total 5.**

Source map: S4 pp.79-81. **Closest past question:** 2025 Q3(b)v-vi. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 1 | Question 4: Ultrasound and optical coherence tomography

4(a) [3 marks]

Question. Explain why an air gap between the ultrasound probe and skin is undesirable and how coupling gel helps.

Expected answer and working. Acoustic impedance $Z=\rho c$ differs greatly between tissue and air. The intensity reflection coefficient at normal incidence is $[(Z2-Z1)/(Z2+Z1)]^2$, close to unity for tissue/air, so little energy enters the body. Gel replaces the air gap with an acoustic coupling layer and improves transmission.

Proposed allocation: Large impedance mismatch (1); Strong reflection / weak transmission (1); Gel displaces air (1). **Total 3.**

Source map: S05 pp.11-15. **Closest past question:** 2025 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [3 marks]

Question. Explain ultrasound speckle and give two acquisition methods for reducing it.

Expected answer and working. Speckle is the granular pattern caused by constructive and destructive interference of echoes from unresolved scatterers. Angular compounding combines views obtained with different beam directions; frequency compounding combines sufficiently different frequencies. The true structures remain aligned while speckle is partly decorrelated. Repeating an identical view does not give the same improvement as averaging independent electronic noise.

Proposed allocation: Coherent interference mechanism (1); Angular compounding (1); Frequency compounding (1). **Total 3.**

Source map: S05 pp.32-35. **Closest past question:** 2024 Q4(a)ii-iii; 2025 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [5 marks]

Question. A small target lies 15 cm deep. Tissue attenuation is 0.80 dB/(cm·MHz). At most 80 dB propagation loss is allowed. Choose among 2, 3 and 5 MHz, ignoring reflection losses. Calculate the loss at each frequency and justify the choice for best resolution. Use $c=1540$ m/s.

Expected answer and working. The pulse travels 30 cm on its round trip. $L=2\cdot0.80\cdot f\cdot15=24f$ dB, so losses are 48, 72 and 120 dB at 2, 3 and 5 MHz. Only 2 and 3 MHz meet the 80 dB budget. Choose 3 MHz: for a fixed number of cycles per pulse, axial resolution improves as 1/f, and focal lateral resolution also improves with smaller wavelength. At 3 MHz $\lambda=1540/(3e6)=0.5133$ mm; 5 MHz is excluded by the loss budget.

Proposed allocation: Round-trip distance (1); Three losses (2); 3 MHz with threshold reasoning (1); Resolution justification (1). **Total 5.**

Source map: S05 pp.16,26-29. **Closest past question:** 2023 Q4(a)iii; 2025 Q4(a)iii-iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [4 marks]

Question. A B-mode array has 192 elements fired in non-overlapping groups of six, one line per group. At 15 cm depth and c=1540 m/s, find the scan lines per frame and maximum frame rate. Assume one pulse per line, sequential lines and no overhead.

Expected answer and working. There are 192/6=32 independent group firings and hence 32 scan lines. Deepest echo time is 2d/c=0.30/1540=194.805 µs. Frame time is 32·194.805 µs=6.23377 ms; maximum frame rate=160.4167 frames/s. Real timing overhead or extra focal-zone transmissions lowers this bound.

Proposed allocation: 32 lines (1); Round-trip time (1); Frame time or correct formula (1); 160.4 frames/s with upper-bound interpretation (1). **Total 4.**

Source map: S05 pp.6-7,17,19,23-24. **Closest past question:** 2024 Q4(a)iv-v. **Status:** Recurring family.

Typical mark-losing error: The number of image rows or depth pixels does not multiply the pulse travel time.

4(e) [4 marks]

Question. Describe the physical cause and image appearance of (i) reverberation and (ii) posterior acoustic enhancement.

Expected answer and working. Reverberation occurs when a pulse reflects repeatedly between strong interfaces (often probe and a nearby strong reflector). Longer travel paths are mapped as deeper echoes, producing repeated, often equally spaced lines. Enhancement occurs behind a region that attenuates less than the surrounding tissue, such as a fluid-filled region; deeper echoes are brighter than expected under the same gain compensation.

Proposed allocation: Reverberation mechanism (1); Repeated-line appearance (1); Low-attenuation path mechanism (1); Brighter posterior region (1). **Total 4.**

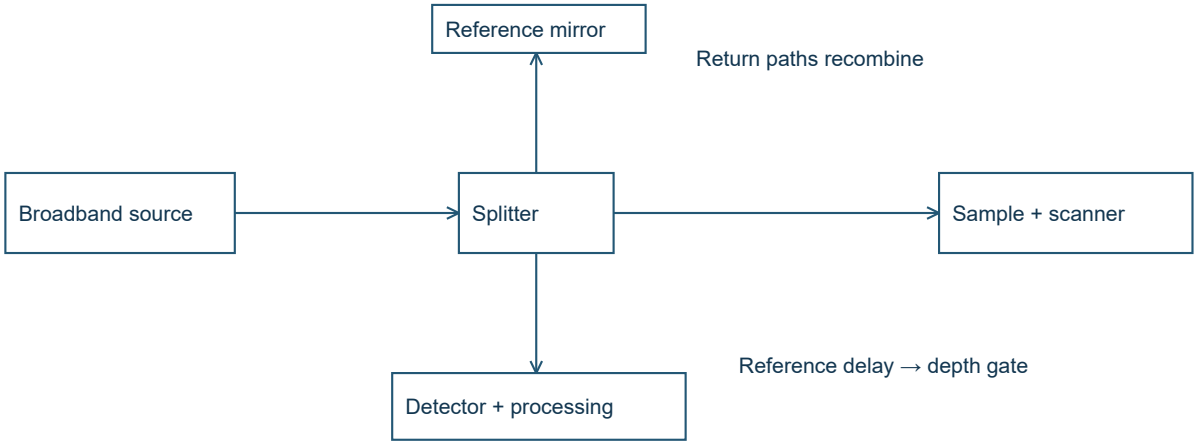
Source map: S05 pp.36-37. **Closest past question:** 2024 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [6 marks]

Question. Draw a basic time-domain OCT interferometer. Explain depth selection and the formation of A- and B-scans. State one medical application and one limitation.

Expected answer and working. A broadband, low-temporal-coherence light source is split into sample and reference arms. Returning sample backscatter and reference-mirror light recombine at a detector. Significant interference envelope occurs when their optical path lengths match within the coherence length. Moving the reference mirror samples optical delay and gives a depth reflectivity profile (A-scan); lateral translation yields successive A-scans forming a cross-sectional B-scan. Retinal-layer imaging is an application. Tissue scattering/absorption limits penetration, so OCT is best for transparent or superficial structures.



Proposed allocation: Source, splitter, two arms, mirror and detector sketch (2); Optical path matching / coherence gate (1); A-scan and B-scan formation (1); Appropriate application (1); Penetration limitation (1). **Total 6.**

Source map: Syllabus p.1; A2 PDF pp.3-7. **Closest past question:** 2024 Q4(b); 2025 Q4(b). **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

DOCUMENT C | SOLUTIONS 2

Worked solutions and proposed marking scheme

Alternative predicted practice paper. These are independent suggested answers and mark allocations, not official university solutions.

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Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 2 | Question 1: Image acquisition and radiotracer kinetics

1(a) [3 marks]

Question. Name three modalities using ionizing radiation and three non-ionizing modalities. For this answer, treat PET and SPECT as distinct modalities.

Expected answer and working. Ionizing examples: projection X-ray, CT and PET (SPECT also accepted). Non-ionizing examples: MRI, ultrasound and OCT (thermal also accepted). Their source energies/interactions differ; the classification is not whether the system requires external power.

Proposed allocation: Three correct ionizing examples (1.5); Three correct non-ionizing examples (1.5). **Total 3.**

Source map: S1 pp.6,29. **Closest past question:** 2023 Q1(a)i; 2024 Q1(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. For CT, MRI, B-mode ultrasound and PET, state the principal physical quantity or signal determining pixel brightness. Include one qualification for MRI or ultrasound.

Expected answer and working. CT estimates linear X-ray attenuation coefficient, commonly expressed as HU relative to water. MRI shows sequence-weighted signal from mobile proton density and relaxation (not simply water concentration). B-mode ultrasound brightness encodes processed echo amplitude/backscatter from tissue interfaces and scatterers, influenced by impedance differences. PET estimates tracer activity distribution after reconstruction and corrections. MRI sequence settings or US gain/processing must be specified for quantitative brightness interpretation.

Proposed allocation: CT attenuation/HU (1); MRI density/relaxation qualification (1); US processed echoes (1); PET activity distribution (1). **Total 4.**

Source map: S1 p.31; S4 p.75; S05 p.19; S06 pp.25-30. **Closest past question:** 2023 Q1(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [3 marks]

Question. Define quantization error and explain two ways to reduce its relative importance in image acquisition.

Expected answer and working. Quantization error is the difference between the analogue input and the represented quantized value. For an ideal rounding ADC it lies within ±half a code step. Increase bit depth to reduce the code step, or amplify the useful signal to occupy the ADC input range without saturation. Adequately dithered oversampling/filtering is another valid approach.

Proposed allocation: Definition (1); Greater bit depth (1); Use input range without clipping or valid oversampling explanation (1). **Total 3.**

Source map: S1 pp.62-72. **Closest past question:** 2023 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [4 marks]

Question. A laboratory needs 18 MBq at 10:00 tomorrow. A sealed tracer stock with physical half-life 4 h is purchased at 18:00 today. Neglect biological clearance. Find the required purchase activity.

Expected answer and working. The elapsed time is 16 h, equivalent to four half-lives. A(t)=A0·2^(-t/T), hence A0=A(t)·2^(t/T)=18·2^4=288 MBq. This is the activity at purchase, before the 16 h decay period; using biological half-life would be inappropriate for a sealed stock.

Proposed allocation: 16 h interval (1); Decay law rearranged (1); Four half-lives / factor 16 (1); 288 MBq (1). **Total 4.**

Source map: S06 pp.6-7. **Closest past question:** 2023 Q1(c)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. Initially there are equal numbers of nuclei of tracers X and Y. Their half-lives are 4 h and 20 h. With equal and constant count-detection efficiencies and no biological clearance, find the earliest time at which X contributes only 25% of their combined count rate.

Expected answer and working. Activity is λN, so AX0/AY0=λX/λY=20/4=5, not 1. The condition AX/(AX+AY)=0.25 gives AX/AY=1/3. The ratio evolves as 5 exp[-(λX-λY)t]. Thus exp[-(ln2)(1/4-1/20)t]=1/15 and t=ln15/[ln2·0.20]=19.53445 h. The fraction is below 25% only after this threshold.

Proposed allocation: Initial activity ratio 5 (1); Fraction becomes ratio 1/3 (1); Ratio decay equation (1); Logarithmic solution (1); 19.53 h / threshold interpretation (1). **Total 5.**

Source map: S06 pp.6-7. **Closest past question:** 2025 Q1(b)iv. **Status:** Recurring family.

Typical mark-losing error: Equal initial nuclei do not mean equal initial count rates.

1(f) [6 marks]

Question. Compare SPECT and PET in terms of emission, direction selection and detection sensitivity. Give one practical limitation of PET.

Expected answer and working. SPECT uses single gamma emission, while PET detects the two 511 keV photons following positron annihilation. SPECT obtains direction information with a physical collimator that rejects most photons; PET uses coincidence-defined LORs. PET can collect a larger fraction of useful events with a detector ring and no comparable in-plane mechanical collimator, and its higher photon energy reduces tissue attenuation. Sensitivity and SNR still depend on acquisition design and correction quality. PET requires reliable short-lived tracer supply/production and expensive equipment; an on-site cyclotron is not a universal requirement for every tracer.

Proposed allocation: Single gamma vs annihilation pair (2); Physical collimator vs coincidence LOR (2); Sensitivity reason (1); Tracer logistics / cost limitation (1). **Total 6.**

Source map: S06 pp.20-29. **Closest past question:** 2024 Q1(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 2: Digital subtraction angiography and CT geometry

2(a) [2 marks]

Question. For a ray through static tissue thickness xt and vessel thickness xv, write pre-contrast intensity I1. Use incident intensity I0 and attenuation coefficients μt and μb; assume monoenergetic radiation and negligible scatter.

Expected answer and working. The exponents add along the ray: I1=I0 exp[-(μt xt+μb xv)]. Here xt excludes the separately specified vessel path, so vessel thickness is not counted twice.

Proposed allocation: Sum of tissue and blood attenuation (1); Correct exponential transmission (1). **Total 2.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [2 marks]

Question. After contrast injection, the vessel coefficient changes from μb to μc while other conditions are unchanged. Write I2.

Expected answer and working. I2=I0 exp[-(μt xt+μc xv)]. The post-contrast vessel coefficient replaces the blood coefficient in this defined simplified model; it is not an extra full blood layer.

Proposed allocation: Replace μb by μc on vessel path (1); Correct common tissue and I0 terms (1). **Total 2.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [3 marks]

Question. Show why I2-I1 does not eliminate the static tissue signal.

Expected answer and working. I2-I1=I0 exp(-μt xt)[exp(-μc xv)-exp(-μb xv)]. The multiplicative factor exp(-μt xt) remains, so the difference still depends on tissue thickness and attenuation. Raw subtraction therefore does not isolate the vessel under this transmission model.

Proposed allocation: Factor common exponential (1); Correct bracket (1); Interpret remaining tissue dependence (1). **Total 3.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [4 marks]

Question. Show that ln(I2)-ln(I1) removes the static tissue term. Evaluate it for μc=1.8 cm^-1, μb=0.2 cm^-1 and xv=0.30 cm. State one acquisition condition required for cancellation.

Expected answer and working. lnI2-lnI1=[lnI0-μt xt-μc xv]-[lnI0-μt xt-μb xv]=-(μc-μb)xv. Thus the value is -(1.8-0.2)·0.30=-0.48, dimensionless. The negative sign follows the requested post-minus-pre convention; reversing it gives positive vessel contrast. Tissue registration/no motion and unchanged exposure normalization are required for cancellation.

Proposed allocation: Expansion of logs (1); Cancellation and signed result formula (1); -0.48 dimensionless (1); Matched motion/exposure condition (1). **Total 4.**

Source map: S2 pp.27-28,48-49. **Closest past question:** 2023 Q2(a)iv. **Status:** Recurring family.

Typical mark-losing error: Preserve the sign specified by the subtraction order.

2(e) [4 marks]

Question. An object has attenuation μ0=0.25 cm^-1 inside a disk of radius R=2.0 cm and zero outside. Derive its parallel-ray projection p(s,θ), where s is the signed perpendicular distance from the disk centre. Evaluate p at s=1.2 cm.

Expected answer and working. For |s|≤R the ray chord has half-length √(R²-s²). Since μ0 is constant, p(s,θ)=2μ0√(R²-s²), independent of θ. Outside the disk it is zero. At s=1.2 cm, chord length=2√(4-1.44)=3.2 cm, so p=0.25·3.2=0.80 (dimensionless). This is a line integral, not transmitted intensity; I/I0=exp(-p).

Proposed allocation: Chord geometry (1); Projection and outside case (1); Angle independence (1); p=0.80 (1). **Total 4.**

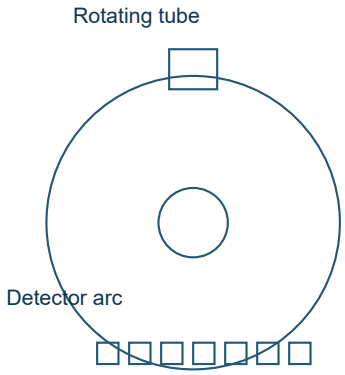
Source map: S3 pp.4,11-16. **Closest past question:** 2023 Q2(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

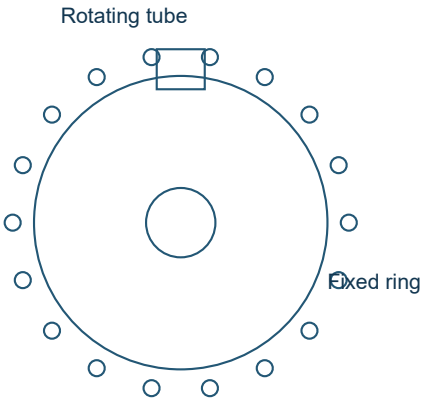
2(f) [6 marks]

Question. Compare third- and fourth-generation CT scanner geometry. Give three distinct features of each, supported by simple labeled sketches.

Expected answer and working. Third generation: the fan-beam X-ray tube and detector arc rotate together; the detector arc covers the fan beam; focused detector collimation is practical, while detector calibration drift can cause rings. Fourth generation: only the tube rotates; a stationary complete detector ring surrounds the patient; detectors can be referenced as the source moves but fixed focused collimation is difficult. Award three valid geometric/performance features per generation; the sketches must show what moves.



3rd: tube + detector arc rotate



4th: tube rotates; ring fixed

Proposed allocation: Third generation three features / diagram (3); Fourth generation three features / diagram (3). **Total 6.**

Source map: S3 pp.37-38,57-58. **Closest past question:** 2023 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(g) [4 marks]

Question. Name four routine CT quality-assurance tests and state what each tests.

Expected answer and working. CT-number accuracy checks known water/air or material HU; uniformity compares the same homogeneous phantom region across the field; spatial resolution assesses small separable structures or an MTF/line pattern; slice-thickness accuracy checks the reconstructed axial thickness against a phantom or prescribed thickness. Image noise is an alternative valid test if linked to a uniform region and repeatable protocol.

Proposed allocation: HU accuracy + purpose (1); Uniformity + purpose (1); Spatial resolution + purpose (1); Slice thickness or noise + purpose (1). **Total 4.**

Source map: S3 pp.31-32 for HU; no explicit QA lecture page supplied. **Closest past question:** 2023 Q2(b)iii. **Status:** Recurring family.

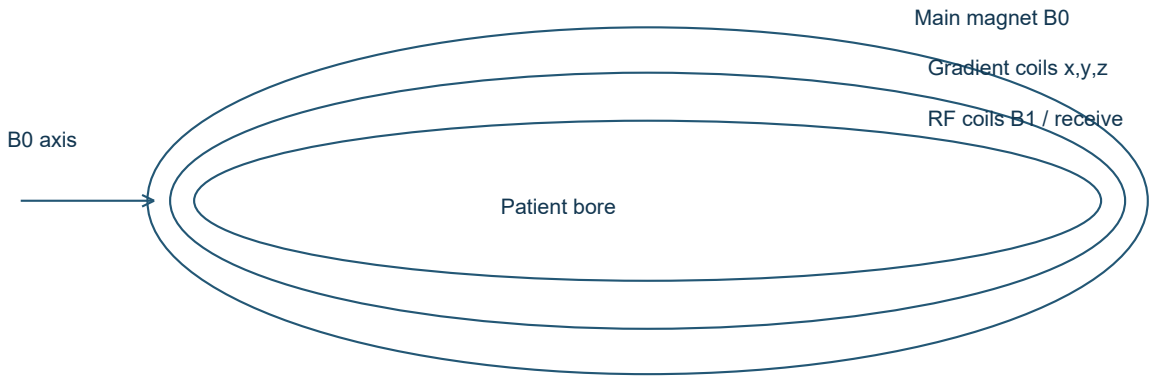
Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 3: MRI hardware and phantom measurements

3(a) [6 marks]

Question. Draw the main components of an MRI scanner and explain the functions of the main magnet, three gradient-coil sets, and RF transmit/receive coils.

Expected answer and working. The main magnet establishes a strong approximately uniform static B0 field. Three independently controlled gradient-coil sets introduce spatial variation in the z-directed field along x, y and z for slice selection and spatial encoding. The RF transmit coil applies B1 at the resonance frequency to tip magnetization; receive coils detect voltage induced by precessing transverse magnetization. The coils surround the patient bore; RF coils lie nearer the subject than the main magnet. RF transmission and reception may use one switched coil or separate coils.



Proposed allocation: Magnet and B0 function (2); Three gradients and spatial encoding (2); RF excitation and induction reception (2). **Total 6.**

Source map: S4 pp.93-101. **Closest past question:** 2023 Q3(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [3 marks]

Question. State three requirements of a main imaging magnet and explain why each matters.

Expected answer and working. High field strength supports spin polarization and signal; spatial homogeneity avoids unintended position-dependent frequency offsets and distortions; temporal stability avoids frequency drift and inconsistent encoding. Adequate patient access or controlled fringe field are also valid requirements if the physical reason is stated.

Proposed allocation: One requirement + reason (1); Second requirement + reason (1); Third requirement + reason (1). **Total 3.**

Source map: S4 p.94. **Closest past question:** 2023 Q3(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [4 marks]

Question. Using the provided high-temperature spin-population approximation, calculate the net proton population imbalance (Nparallel-Nantiparallel)/Ntotal as a percentage at B0=3.0 T and 27°C. Explain why the phantom volume is unnecessary.

Expected answer and working. Convert T=27+273.15=300.15 K. Fraction δ=γhB0/(4πkT)=(267.52e6·6.63e-34·3)/(4π·1.38e-23·300.15)=1.0222656e-5. Percentage=0.00102227%. This is the small excess parallel population over the antiparallel population, not the total fraction in the parallel state. Since Ntotal divides out, neither volume nor number of water molecules is needed for this fraction.

Proposed allocation: Kelvin conversion (1); Fraction equation and substitution (1); 0.001022% with correct scale (1); Population difference and volume cancellation (1). **Total 4.**

Source map: S4 pp.12-17. **Closest past question:** 2023 Q3(b)i. **Status:** Recurring family.

Typical mark-losing error: Multiply the population fraction by 100 for percent; it is not 0.1% or the entire parallel population.

3(d) [4 marks]

Question. A central 5.0 mm thick slice is selected with Gz=0.40 mT/cm at 3.0 T. Calculate the full RF bandwidth and absolute frequency endpoints.

Expected answer and working. Gz=0.40e-3/0.01=0.040 T/m. Full ΔB=GzΔz=0.040·0.0050=2.0e-4 T. With γbar=γ/(2π), Δf=8515.4261 Hz. f0=127.73139113 MHz and half-bandwidth=4257.7130 Hz. Endpoints are 127,727,133.41 and 127,735,648.84 Hz. The slice is centred at the magnet isocentre; no spatial offset correction is needed.

Proposed allocation: Gradient conversion (1); Full bandwidth relation (1); 8515 Hz (1); Correct endpoints (1). **Total 4.**

Source map: S4 pp.15,48-54. **Closest past question:** 2023 Q3(b)ii-iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. A conventional single-slice acquisition has 240 readout samples and 160 phase-encoding steps. TR=1.25 s and two averages are acquired. Calculate the scan time. If only readout samples are doubled, explain the effect on the ideal scan time under the same TR.

Expected answer and working. Tscan=Nphase·TR·NEX=160·1.25·2=400 s=6 min 40 s. The readout sample count is collected within each readout window, whereas the phase dimension requires repetitions. Under the explicit assumption that the doubled readout still fits within the same TR, doubling readout samples does not change the ideal 400 s scan time. It can alter bandwidth/readout duration and resolution, so the unchanged-TR assumption matters.

Proposed allocation: Nphase not Nread (1); Include two averages (1); 400 s / 6 min 40 s (1); Readout-vs-repetition explanation (1). **Total 4.**

Source map: S4 pp.61-62,71,92. **Closest past question:** 2025 Q3(b)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [4 marks]

Question. Give two phantom-based MRI checks and explain how an appropriate phantom feature would support each measurement.

Expected answer and working. A homogeneous water region supports B0 homogeneity/frequency uniformity measurements or image uniformity assessment under fixed coil conditions. A phantom with known thin ramps or geometry can assess slice thickness or geometric accuracy. Alternatively, known point/line spacings assess spatial resolution, and repeat images or a valid noise method assess SNR. A plain sphere by itself does not automatically provide all these calibrations; the feature and acquisition must match the metric.

Proposed allocation: First valid metric (1); Suitable phantom feature/method (1); Second valid metric (1); Suitable feature/method (1). **Total 4.**

Source map: S4 pp.52-54,92,94,98; phantom-check design is applied context. **Closest past question:** 2023 Q3(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 2 | Question 4: Ultrasound and thermal imaging

4(a) [3 marks]

Question. State three advantages and three limitations of ultrasound compared with CT or MRI.

Expected answer and working. Advantages: portability, real-time high temporal resolution, and no ionizing radiation. Limitations: operator dependence, restricted acoustic windows/field of view, and difficulty imaging through bone or air. Award 0.5 for each distinct correct item; cost is an acceptable alternative advantage.

Proposed allocation: Three advantages (1.5); Three limitations (1.5). **Total 3.**

Source map: S05 pp.2-5. **Closest past question:** 2023 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [6 marks]

Question. Explain reflection, scattering and absorption of ultrasound in tissue. Link each to an image consequence.

Expected answer and working. Reflection occurs at impedance discontinuities and creates boundary echoes; a strong smooth interface can redirect energy away from the probe. Scattering from small/irregular structures distributes sound in many directions; returning components provide tissue texture and coherent interference generates speckle. Absorption converts acoustic energy to heat and reduces echo amplitude with path length, contributing to loss of deep signal and posterior shadowing behind highly attenuating structures.

Proposed allocation: Reflection mechanism and image effect (2); Scattering mechanism and image effect (2); Absorption mechanism and image effect (2). **Total 6.**

Source map: S05 pp.13-16,35-38. **Closest past question:** 2023 Q4(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [5 marks]

Question. A target lies 18 cm deep in tissue with attenuation 0.70 dB/(cm·MHz). The receiver accepts at most 95 dB propagation loss. Choose the highest usable frequency from 2, 3 and 4 MHz. Justify using calculated losses and the resolution trade-off.

Expected answer and working. Round-trip loss is $L=2\alpha f d=2\cdot 0.70\cdot 18\cdot f=25.2f$ dB. For 2, 3 and 4 MHz, losses are 50.4, 75.6 and 100.8 dB. Frequencies 2 and 3 MHz are usable; choose 3 MHz for improved resolution at a fixed pulse-cycle count and focus geometry. Four MHz exceeds 95 dB, while choosing 2 MHz would unnecessarily sacrifice resolution under the stated assumptions.

Proposed allocation: Round trip in loss expression (1); All three losses (2); Select 3 MHz (1); Explain resolution/penetration balance (1). **Total 5.**

Source map: S05 pp.16,28-29. **Closest past question:** 2023 Q4(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [3 marks]

Question. At 18 cm imaging depth and c=1500 m/s, the required B-mode frame rate is 50 frames/s. Find the maximum integer number of sequential scan lines per frame, ignoring overhead.

Expected answer and working. A pulse echo needs tline=2·0.18/1500=0.000240 s. One frame is 1/50=0.020 s. Nlines≤0.020/0.000240=83.333, so at most 83 complete lines can be acquired. Round down: 84 lines would exceed the frame period.

Proposed allocation: 240 μs line time (1); Ratio gives 83.333 (1); 83 integer lines with reason (1). **Total 3.**

Source map: S05 pp.6-7,17,19. **Closest past question:** 2023 Q4(a)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [4 marks]

Question. With a labeled block diagram, explain how an uncooled thermal camera forms a medical thermogram.

Expected answer and working. Warm skin emits infrared radiation. IR-transmitting optics focus it onto a focal-plane detector array. In a microbolometer, absorbed radiation raises detector temperature and changes its electrical resistance. Readout electronics and an ADC digitize the signal; calibration/nonuniformity correction and emissivity assumptions support conversion to a displayed surface-temperature map. The map is not a direct image of deep anatomy.



Proposed allocation: Emitted IR and optics (1); Array/microbolometer conversion (1); Readout/ADC and calibrated processing (1); Surface-temperature display with labels (1). **Total 4.**

Source map: A1 PDF pp.3-5; syllabus p.1. **Closest past question:** 2023 Q4(b)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [2 marks]

Question. State two appropriate medical uses of thermal imaging and link each to the physiological information observed.

Expected answer and working. Diabetic-foot follow-up can monitor local temperature asymmetry associated with inflammation or altered perfusion. Burn/wound monitoring can compare surface-temperature patterns related to perfusion and healing. Other justified uses include inflammatory-joint monitoring or controlled surface-temperature screening. A thermal pattern is nonspecific and requires clinical context.

Proposed allocation: First use + observed physiology (1); Second use + observed physiology (1). **Total 2.**

Source map: A1 PDF pp.5-7. **Closest past question:** 2023 Q4(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(g) [2 marks]

Question. Give two advantages of thermal imaging and two acquisition controls needed for reliable comparisons over time.

Expected answer and working. Advantages include noncontact measurement and no ionizing radiation, enabling repeated observations. Controls include stable ambient conditions/patient acclimatization and consistent camera calibration, emissivity, distance and view. Credit two distinct controls; changes in room conditions can mimic physiological change.

Proposed allocation: Two advantages (1); Two acquisition controls (1). **Total 2.**

Source map: A1 PDF pp.3,8-9. **Closest past question:** 2023 Q4(b)iii; controls are assignment coverage. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

DOCUMENT C | SOLUTIONS 3

Worked solutions and proposed marking scheme

Challenging application-based practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 3 | Question 1: Image quality and quantitative nuclear imaging

1(a) [4 marks]

Question. Two systems have normalized Gaussian PSFs. System A has σx=0.40 mm and σy=1.00 mm; B has σx=σy=0.70 mm. Calculate each directional FWHM and choose the better system for separating structures 1.2 mm apart along x, using separation greater than FWHM as the stated criterion.

Expected answer and working. FWHM=2.35482σ. For A, widths are 0.94193 mm in x and 2.35482 mm in y. For B, both widths are 1.64837 mm. At 1.2 mm separation along x, A satisfies the stated criterion while B does not. A is anisotropic and is worse than B in y, so it is not universally superior; orientation matters.

Proposed allocation: A widths (1); B widths (1); A chosen under stated criterion (1); Directional limitation (1). **Total 4.**

Source map: S1 pp.39-48. **Closest past question:** 2024 Q1(a)ii-iii; 2025 Q1(a). **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. Two regions have mean signals 120 and 100 units and noise standard deviation 5 units. Compute their CNR. A small lesion voxel contains 25% of the first region and 75% of the second. Find its mean signal and CNR against the second region, assuming unchanged noise.

Expected answer and working. For pure regions, CNR=(120-100)/5=4. The mixed voxel signal is 0.25·120+0.75·100=105 units. Against the 100-unit background its CNR is (105-100)/5=1. Partial volume reduces apparent contrast even when receiver noise is unchanged, so detectability deteriorates.

Proposed allocation: CNR=4 (1); Mixture formula (1); 105 units (1); CNR=1 with partial-volume explanation (1). **Total 4.**

Source map: S1 pp.55-56. **Closest past question:** 2024 Q2(a)iii is closest contrast precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [2 marks]

Question. An image with independent zero-mean noise has SNR=12. How many total independent acquisitions must be averaged to reach SNR=30? Explain why identical-view ultrasound speckle need not obey this scaling.

Expected answer and working. SNRN=SNR1√N gives N=(30/12)²=6.25. At least 7 whole acquisitions are needed (six would give only 29.39). Identical-view speckle is coherent and correlated, so its pattern does not average away as independent noise would.

Proposed allocation: Seven total acquisitions from square law (1); Speckle correlation explanation (1). **Total 2.**

Source map: S1 pp.52-53; S05 pp.32-35. **Closest past question:** 2024 Q4(a)iii is closest compounding precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [6 marks]

Question. A tracer has physical half-life 8 h and biological half-life 12 h. Its initial in-body activity is 160 MBq. Calculate the effective half-life, activity after 6 h, and time at which activity first falls to 20 MBq. Explain the role of independent clearance.

Expected answer and working. λeff=λphys+λbio, hence 1/Teff=1/8+1/12=5/24 h^-1 and Teff=4.8 h. After 6 h, A=160·2^(-6/4.8)=67.27171 MBq. Reaching 20 MBq is a factor-of-eight decrease, or three effective half-lives: t=3·4.8=14.4 h. Independence and exponential clearance allow the survival factors to multiply and the rate constants to add; effective half-life is shorter than either component.

Proposed allocation: Sum-rate law and assumption (1); 4.8 h (1); 6 h decay substitution (1); 67.27 MBq (1); Three half-lives (1); 14.4 h with interpretation (1). **Total 6.**

Source map: S06 pp.6-8. **Closest past question:** 2024 Q1(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [5 marks]

Question. A non-paralyzable gamma-camera channel records n=40,000 counts/s. Its dead time is τ=5.0 μs. Using τ=(N-n)/(nN), derive N in terms of n and τ, calculate the true rate N and the fraction of true events lost.

Expected answer and working. Rearrange nNτ=N-n to N(1-nτ)=n, hence N=n/(1-nτ). Here nτ=40000·5e-6=0.20, so N=40000/0.80=50,000 counts/s. Fraction lost=(N-n)/N=10000/50000=0.20, or 20%. The formula requires nτ<1; its measured-rate saturation limit is 1/τ=200,000 counts/s. A paralyzable model would not use this same correction.

Proposed allocation: Algebraic rearrangement (2); Unit substitution and N=50,000 (1); Lost fraction=20% (1); Validity / model interpretation (1). **Total 5.**

Source map: S06 pp.14-17. **Closest past question:** No direct historical dead-time question. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [4 marks]

Question. Describe true, random and scattered coincidences in PET. Explain one way that false coincidences affect the reconstructed image.

Expected answer and working. A true coincidence records an unscattered pair from the same annihilation, defining the appropriate LOR. A random coincidence records photons from different annihilations inside the acceptance window. A scattered coincidence includes a photon deflected before detection, misplacing the inferred LOR even if it belongs to the same annihilation. Random/scattered events add incorrectly located counts, lowering contrast and quantitative accuracy unless corrected.

Proposed allocation: True definition (1); Random definition (1); Scattered definition (1); Image consequence (1). **Total 4.**

Source map: S06 pp.32-34. **Closest past question:** 2025 Q1(b)ii is closest coincidence precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 2: Radiographic design and reconstruction constraints

2(a) [4 marks]

Question. An X-ray target has actual projected electron-track width F=2.0 mm and bevel angle θ=12°. Calculate effective focal spot f=F sinθ and beam coverage 2D tanθ at D=80 cm. Recalculate at 8° and explain the image/FOV trade-off.

Expected answer and working. At 12°, f=2 sin12°=0.41582 mm; coverage=160 tan12°=34.0090 cm. At 8°, f=0.27835 mm and coverage=22.4865 cm. The smaller bevel angle reduces effective spot and geometric blur but narrows the field available at the same source distance. These projected-spot calculations hold F constant and do not themselves change the actual electron-track heating area.

Proposed allocation: 12° spot and coverage (1); 8° spot and coverage (1); Improved geometric resolution (1); Reduced coverage / fixed-F interpretation (1). **Total 4.**

Source map: S2 pp.9-13. **Closest past question:** 2025 Q2(a)ii-iii. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [4 marks]

Question. Without a grid, a detector receives 600 primary and 400 scattered photons in a stated interval. A grid transmits 70% of primary photons and 10% of scattered photons. Find the detected total and primary fraction with the grid. If output is scaled to restore the original total detected count, estimate the relative exposure factor; assume counts scale linearly with exposure.

Expected answer and working. Grid counts are P=0.70·600=420 and S=0.10·400=40, total 460. Primary fraction rises from 600/1000=60% to 420/460=91.304%. To restore 1000 total counts requires a factor 1000/460=2.17391 in exposure under the stated linear model. Scatter suppression can improve contrast but maintaining detector counts increases patient exposure. This simplified count-matching factor is not a universal grid Bucky factor.

Proposed allocation: 420 primary + 40 scatter (1); 460 total (1); 91.3% primary (1); 2.174 exposure factor and trade-off (1). **Total 4.**

Source map: S2 pp.34-38. **Closest past question:** 2025 Q2(a)i is closest hardware precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [5 marks]

Question. A CT scanner measures air as -1040 HU and water as +20 HU. A linear calibration is h=a hm+b. Determine a and b using true air=-1000 HU and water=0 HU. Correct a measured tissue value of 232 HU, then compute its μ if μwater=0.200 cm^-1.

Expected answer and working. The constraints are -1000=-1040a+b and 0=20a+b. Subtracting gives 1000=1060a, so a=50/53=0.94339623 and b=-1000/53=-18.86792453 HU. For hm=232, h=(50·232-1000)/53=200 HU. HU=1000(μ-μwater)/μwater gives μ=μwater(1+h/1000)=0.200·1.2=0.240 cm^-1. The corrected value, not the raw measured value, enters the attenuation conversion.

Proposed allocation: Two calibration equations (1); a (1); b (1); Corrected 200 HU (1); μ=0.240 cm^-1 (1). **Total 5.**

Source map: S3 pp.31-32. **Closest past question:** 2025 Q2(b)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [8 marks]

Question. For a 2×2 image [[a,b],[c,d]], unit-weight projections are top row=8, bottom row=12, left column=7 and right column=13. Start from zero and use one equal-residual ART sweep in that order. Then an extra isolated ray gives a=2. Determine the unique image satisfying all data and verify it. Distinguish this final constrained solution from the one-sweep estimate.

Expected answer and working. From zero, top update gives [[4,4],[0,0]], then bottom gives [[4,4],[6,6]]. Left predicted sum=10, residual=-3; subtract 1.5 from a,c to obtain [[2.5,4],[4.5,6]]. Right predicted sum=10, residual=+3; add 1.5 to b,d, yielding [[2.5,5.5],[4.5,7.5]]. This fits the row/column data. The new independent constraint sets a=2. Then b=8-2=6; c=7-2=5; d=12-5=7. Thus the unique constrained image is [[2,6],[5,7]], with right sum 6+7=13. Simply setting a=2 in the one-sweep estimate would break existing sums; the other entries must change too. The direct constrained solution is not claimed to be a single extra ART ray update.

Proposed allocation: Top and bottom intermediate images (2); Left residual and update (1); Right residual and one-sweep estimate (1); Use isolated a=2 (1); b=6,c=5,d=7 (1); Verify all projections (1); Distinguish constrained solution from one-sweep ART (1). **Total 8.**

Source map: S3 pp.26-28. **Closest past question:** 2024 Q2(b)iii. **Status:** Original application of ART.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(e) [4 marks]

Question. Explain why noisy or incomplete projections can make exact ray-sum agreement undesirable or impossible. Give one advantage and one limitation of iterative reconstruction relative to direct analytic reconstruction.

Expected answer and working. Noise makes measured rays mutually inconsistent with a single ideal attenuation image, and incomplete views leave unconstrained object features. Iteration can include a forward model of acquisition and seek an approximate statistically appropriate fit, rather than satisfy every noisy ray exactly. It can accommodate incomplete geometry or detector characteristics. Its computational cost is greater and results depend on model/constraints and stopping; uncontrolled ART can oscillate with inconsistent noisy data. Direct analytic methods are fast but usually assume an ideal, adequately sampled geometry.

Proposed allocation: Noisy inconsistency (1); Incomplete-data ambiguity (1); Iterative modeling advantage (1); Computation/stability limitation vs analytic (1). **Total 4.**

Source map: S3 pp.14,26-29. **Closest past question:** 2025 Q2(c)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 3: MRI signal optimization and acquisition design

3(a) [4 marks]

Question. For a spoiled GE sequence, explain the roles of TR, TE, flip angle α and T2*. State which may be chosen by the operator and which is a tissue/field property.

Expected answer and working. TR is the time between successive excitations of the same slice and controls longitudinal recovery. TE is excitation-to-echo-centre time and controls transverse decay weighting. Flip angle is the RF-induced magnetization rotation and affects excitation and steady-state saturation. T2* describes transverse decay including intrinsic T2 and field inhomogeneity. TR, TE and α are acquisition choices; T2* depends on tissue and the local/static field conditions.

Proposed allocation: TR (1); TE (1); α (1); T2* with classification (1). **Total 4.**

Source map: S4 pp.22,29-38,62,78. **Closest past question:** 2024 Q3(b)iii-v. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [5 marks]

Question. For a fixed-TR, fixed-TE spoiled GE sequence with E=exp(-TR/T1), signal is proportional to (1-E)sinα/(1-E cosα). Derive the maximizing angle for $0<\alpha<\pi/2$. Evaluate it for TR=0.40 s and T1=1.00 s.

Expected answer and working. Treat E as constant. Differentiating the angle-dependent fraction gives $(1-E)[\cos\alpha(1-E\cos\alpha)-E\sin^2\alpha]/(1-E\cos\alpha)^2 = (1-E)(\cos\alpha-E)/(1-E\cos\alpha)^2$. Set the numerator to zero: cosα=E. The derivative changes from positive to negative across this point, so it is a maximum. With E=e^-0.4=0.67032005, αE=acos(E)=47.9082°. The result assumes spoiled transverse magnetization and constant noise/other acquisition settings.

Proposed allocation: Correct quotient differentiation (2); Simplification cosα-E (1); Stationary point and maximum (1); 47.91° (1). **Total 5.**

Source map: S4 p.78. **Closest past question:** 2024 Q3(b)iv-v. **Status:** Original derivation of supplied lecture equation.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [5 marks]

Question. At TR=0.40 s, TE=0.012 s and T1=1.00 s, compare two otherwise identical regions with T2*=0.040 s and 0.080 s. Use the Ernst angle from (b). Calculate their normalized GE signals for ρ=1, their signal ratio, and the percentage increase for the longer-T2* region.

Expected answer and working. At the Ernst angle the recovery/flip factor is √[(1-E)/(1+E)]=√[(1-e^-0.4)/(1+e^-0.4)]=0.44426942. For T2*=0.040 s, S1=0.44426942 exp(-0.012/0.040)=0.32912288. For 0.080 s, S2=0.44426942 exp(-0.012/0.080)=0.38238624. Their ratio S2/S1=exp(0.15)=1.16183424, so the second signal is 16.1834% larger. The same density/T1 does not imply equal GE intensity because transverse decay differs.

Proposed allocation: Common steady-state factor (1); S1 calculation (1); S2 calculation (1); Ratio 1.16183 (1); 16.18% increase and interpretation (1). **Total 5.**

Source map: S4 pp.75,78. **Closest past question:** 2024 Q3(b)v. **Status:** Original multi-tissue variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [4 marks]

Question. A 256×128 readout-by-phase image uses TR=0.40 s and three averages. Calculate total scan time. If the phase steps are doubled and averages reduced to one, find the new scan time and explain the change.

Expected answer and working. Original time=128·0.40·3=153.6 s. New time=256·0.40·1=102.4 s. The new/original time ratio is 2/3. Each phase step needs a repetition; reducing three averages to one more than offsets doubling phase steps. The readout dimension is not a separate repetition multiplier under this conventional sequence model.

Proposed allocation: Original formula and time (1); New formula and time (1); Ratio 2/3 (1); Repetition explanation (1). **Total 4.**

Source map: S4 pp.61-62,71,92. **Closest past question:** 2025 Q3(b)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [4 marks]

Question. Each complete slice sequence occupies 25 ms within TR=0.40 s. Find the maximum ideal interleaved slice count and the scan time for 12 slices using the original settings of (d). Explain the required frequency and storage changes.

Expected answer and working. There are floor(400/25)=16 full slice slots per TR. Twelve slices fit, so each receives one excitation per 0.40 s and all 12 complete the original 128 phase steps and three averages in 153.6 s, neglecting preparation/overhead. Change RF centre frequency for slice position, retain the prescribed bandwidth/gradient for thickness, and store each slice's k-space separately. Multiplying 153.6 by 12 would describe serial rather than interleaved acquisition.

Proposed allocation: 16 slots (1); 12 fit /153.6 s (1); RF centre frequency (1); Separate k-space and same per-slice TR (1). **Total 4.**

Source map: S4 pp.72-74. **Closest past question:** 2024 Q3(b)ii; 2025 Q3(b)iv. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [3 marks]

Question. Under the approximate scaling SNR proportional to voxel volume times √(total ADC-on time), what happens to SNR if voxel volume is halved with unchanged ADC-on time? How much additional acquisition time would restore the original SNR? State a practical cost.

Expected answer and working. Halving voxel volume halves SNR at the same total ADC-on time under the stated model. To recover a factor of two in SNR needs a factor of four in acquisition time because √4=2. The longer scan increases motion risk and reduces temporal resolution; signal averaging cannot fully remove motion or correlated artifacts.

Proposed allocation: SNR halves (1); Fourfold time (1); Practical cost (1). **Total 3.**

Source map: S4 p.92; S1 pp.52-53. **Closest past question:** No direct historical voxel-time trade-off question. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 3 | Question 4: Ultrasound design and optical ranging

4(a) [6 marks]

Question. A target reflects 50% of incident intensity and lies in tissue with attenuation 0.60 dB/(cm·MHz). At f=3 MHz the total permissible echo-intensity loss is 90 dB, measured relative to launched intensity. Calculate the maximum depth, accounting for reflection and round-trip propagation. Ignore all other losses.

Expected answer and working. Reflection contributes a positive loss Lr=-10 log10(0.50)=3.01030 dB. The propagation budget is 90-3.01030=86.98970 dB. For one-way depth d in cm, Lprop=2·0.60·3·d=3.6d dB. Therefore dmax=86.98970/3.6=24.16381 cm. Check: 2·0.60·3·24.16381+3.01030=90 dB. A pressure-amplitude coefficient would require a different dB conversion; here intensity reflectance is explicitly given.

Proposed allocation: Intensity reflection loss with 10 log (1); 3.0103 dB (1); Remaining propagation budget (1); Round-trip attenuation expression (1); 24.16 cm (1); Check / physical interpretation (1). **Total 6.**

Source map: S05 pp.15-16,28; local discussion PDF p.5. **Closest past question:** 2025 Q4(a)iv is closest depth precedent. **Status:** Original application with reflection loss.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [5 marks]

Question. At 3 MHz and c=1540 m/s, the pulse contains three cycles. The focal distance is 60 mm and aperture width 20 mm. Compute wavelength, axial resolution and focal lateral resolution using the supplied lecture approximations. What would doubling frequency do if all other values remained fixed?

Expected answer and working. λ=c/f=1540/(3e6)=0.000513333 m=0.513333 mm. Axial resolution=Ncλ/2=3·0.513333/2=0.770 mm. F-number=60/20=3; lateral resolution≈λF/L=0.513333·3=1.540 mm. Doubling frequency halves both numerical resolution distances, improving detail, but doubles propagation loss per cm in the given attenuation model and reduces accessible depth.

Proposed allocation: Wavelength (1); Axial 0.770 mm (1); F-number 3 (1); Lateral 1.540 mm (1); Frequency trade-off (1). **Total 5.**

Source map: S05 pp.25-29. **Closest past question:** 2025 Q4(a)iii is closest resolution precedent. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [4 marks]

Question. A 128-line frame images to 20 cm at c=1540 m/s. Compare maximum frame rates for one transmit focus per line and three sequential transmit foci per line. Assume no parallel reception or timing overhead.

Expected answer and working. One pulse echo requires 0.40/1540=259.740 μs. For 128 lines and one focus, frame time=128·0.40/1540=0.0332468 s, so FR=30.078125 frames/s. Three transmit foci require three firings/line, so time triples to 0.0997403 s and FR=10.02604 frames/s. More transmit focal zones improve focusing coverage at the expense of temporal resolution.

Proposed allocation: Round-trip echo time (1); 30.08 frames/s (1); Threefold time /10.03 frames/s (1); Focusing-time trade-off (1). **Total 4.**

Source map: S05 pp.6-7,19,25-29. **Closest past question:** 2024 Q4(a)v-vi. **Status:** Original application of timing.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [4 marks]

Question. Explain angular and frequency compounding for speckle reduction. State why improved smoothness can conceal a loss of temporal resolution.

Expected answer and working. Angular compounding combines registered views acquired with different beam directions; frequency compounding combines views at distinct frequencies within probe bandwidth. Speckle changes because coherent phase combinations differ, while anatomical structure remains more consistent. Acquiring several views for one displayed composite can reduce the number of independent composite frames per second or introduce temporal averaging/motion blur. A smooth display alone does not prove better spatial or temporal resolution.

Proposed allocation: Angular mechanism (1); Frequency mechanism (1); Decorrelated speckle / consistent anatomy (1); Acquisition-time / motion trade-off (1). **Total 4.**

Source map: S05 pp.32-35. **Closest past question:** 2024 Q4(a)iii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [6 marks]

Question. An OCT source has Gaussian spectrum with centre wavelength $\lambda_0=840$ nm and FWHM bandwidth $\Delta\lambda=60$ nm. Using $\delta z_{air}=(2\ln 2/\pi)\lambda_0^2/\Delta\lambda$, find axial resolution in air and in tissue of refractive index $n=1.40$, taking $\delta z_{tissue}=\delta z_{air}/n$. Explain why a broader spectrum improves axial resolution and why this calculation does not determine lateral resolution.

Expected answer and working. Use consistent wavelengths: $\lambda_0^2/\Delta\lambda=840^2/60=11760$ nm. Multiplying by $2\ln 2/\pi=0.44127120$ gives $\delta z_{air}=5189.3495$ nm$=5.18935$ μm. In tissue $\delta z=5.18935/1.40=3.70668$ μm. A broader spectrum has a shorter coherence gate, separating reflections at smaller optical delays. Lateral resolution depends on focusing optics/beam spot rather than this temporal-coherence formula. This is assignment-level OCT coverage; it is not a recurring historical calculation.

Proposed allocation: Formula and consistent units (1); 5.189 μm in air (1); 3.707 μm in tissue (1); Broad bandwidth shortens coherence gate (1); Lateral focusing distinction (1); Physical interpretation / correctly distinguishes optical and geometric depth (1). **Total 6.**

Source map: A2 PDF p.5; syllabus p.1; Aumann technical chapter cited in A2. **Closest past question:** No direct historical OCT-resolution numerical. **Status:** Original assignment-supported coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

DOCUMENT C | SOLUTIONS 4

Worked solutions and proposed marking scheme

Full-syllabus coverage practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 4 | Question 1: Sampling, image processing and nuclear instrumentation

1(a) [5 marks]

Question. An ideal 10-bit rounding ADC has input range 0-2.048 V and uses 1024 equal-width bins. Calculate its code width and maximum quantization-error magnitude. A useful input signal spans only 0-0.128 V; propose a gain to use the full range and explain the risk. Give three ADC specifications relevant to imaging.

Expected answer and working. Bin width=2.048/1024=0.002 V=2 mV; maximum rounding error is half a bin, 1 mV. Nominal gain=2.048/0.128=16, but practical gain should leave headroom because an unexpectedly larger input clips. Relevant specifications include bit depth/dynamic range, permitted input voltage range and maximum sampling rate (or usable input bandwidth). Bin endpoints are defined in the question, avoiding the 2^n versus 2^n-1 convention ambiguity.

Proposed allocation: 2 mV width (1); 1 mV error bound (1); Gain 16 (1); Clipping/headroom (1); Three relevant specifications (1). **Total 5.**

Source map: S1 pp.62-68. **Closest past question:** 2023 Q1(b)i-iii. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. A real low-pass analogue signal contains frequencies up to 18 kHz. State the theoretical Nyquist sampling bound and explain the role of an analogue anti-alias filter. A 16-fold oversampling/decimation scheme is then used; what ideal equivalent-bit improvement is suggested by the lecture, and under what noise assumption?

Expected answer and working. The theoretical bound is $f_s \geq 2 \cdot 18 = 36$ ksample/s; a practical design uses margin and an appropriate filter transition band. The analogue anti-alias filter suppresses out-of-band components before sampling so they cannot fold into the signal band. The lecture rule is one effective bit per fourfold oversampling, hence $16 = 4^2$ gives +2 bits after low-pass filtering/decimation. It assumes adequately uncorrelated/noise-like quantization errors; oversampling alone cannot guarantee extra bits for a deterministic undithered error.

Proposed allocation: 36 ksample/s bound (1); Pre-ADC filtering / alias explanation (1); +2 bits (1); Noise assumption / decimation condition (1). **Total 4.**

Source map: S1 pp.67-72. **Closest past question:** No direct historical sampling numerical. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [3 marks]

Question. Compare spatial low-pass and high-pass filtering in terms of detail, noise and diagnostic visibility.

Expected answer and working. Low-pass filtering suppresses high spatial frequencies, reducing noise but smoothing edges and small features. High-pass filtering emphasizes rapid spatial changes and edges but can increase visible noise. Neither guarantees greater diagnostic utility: detectability depends on lesion contrast, resolution and noise together, and processing cannot restore genuinely absent measurements without assumptions.

Proposed allocation: Low-pass effect and trade-off (1); High-pass effect and trade-off (1); Visibility depends on task / information limit (1). **Total 3.**

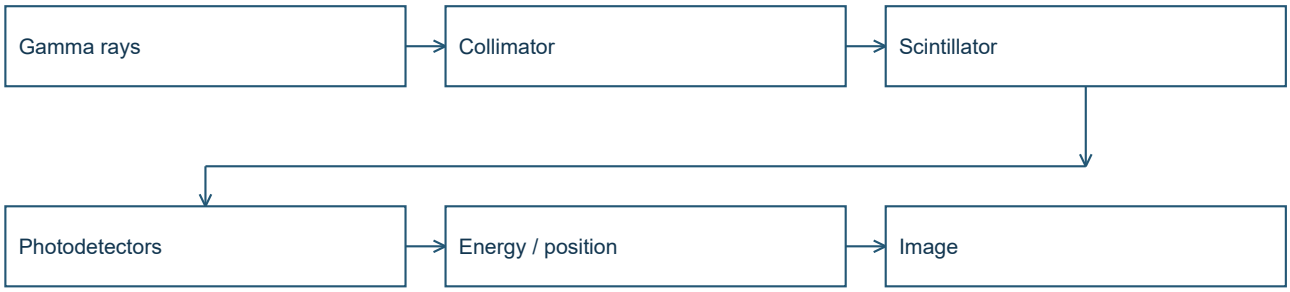
Source map: S1 pp.33-37,73-74. **Closest past question:** No direct historical filtering question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [5 marks]

Question. Draw the signal path of a scintillation gamma camera. Explain the roles of collimator, scintillator, photodetectors, position/energy electronics and energy-window selection.

Expected answer and working. The collimator admits a restricted set of photon directions at a major sensitivity cost. Absorption in a scintillator creates light; photodetectors convert and amplify that light. Relative photodetector outputs locate the event, and their total/pulse height estimates deposited energy. An energy window around the photopeak rejects many scattered lower-energy events. Accepted positions accumulate into an image of tracer distribution. The chain is gamma photon → collimator → scintillator → photodetectors → position/energy logic → energy acceptance → image.



Proposed allocation: Collimator direction selection (1); Scintillator conversion (1); Photodetection (1); Position/energy determination (1); Window selection and labeled chain (1). **Total 5.**

Source map: S06 pp.11-13,21,35-37. **Closest past question:** 2024 Q1(b)i is closest SPECT context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [4 marks]

Question. Explain why SPECT needs scatter and attenuation correction. State how a combined SPECT/CT system assists quantitative interpretation.

Expected answer and working. Scatter changes photon direction and can deposit a count at an incorrect location; energy-window measurements help estimate/reject its contribution. Attenuation reduces counts more strongly for photons from deeper locations, so raw image brightness is not proportional to local activity alone. CT supplies a patient-specific attenuation map, appropriately converted to the gamma energy, and anatomical localization for the functional SPECT distribution. Registration errors can still impair the fusion.

Proposed allocation: Scatter mechanism (1); Energy-window approach (1); Depth-dependent attenuation (1); CT attenuation/localization benefit (1). **Total 4.**

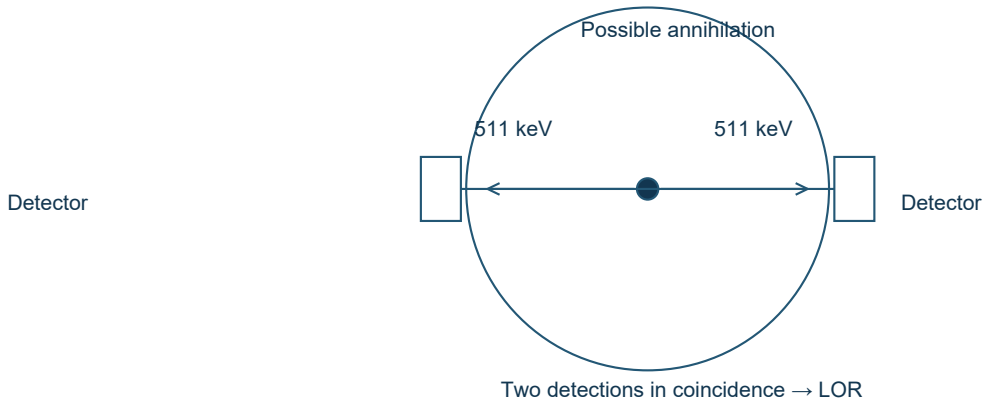
Source map: S06 pp.20-24. **Closest past question:** 2024 Q1(b)ii is closest SPECT/PET comparison. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [4 marks]

Question. Using a simple sketch, distinguish a PET line of response from the position of an annihilation event. Explain why the detection of one coincidence does not by itself create a complete image.

Expected answer and working. Two opposing detectors define a chord/LOR containing the annihilation position under ideal unscattered geometry. Ordinary coincidence establishes that the pair belongs together but not a unique position along the chord. Multiple events and many LOR orientations constrain the tracer distribution tomographically. Counting noise, scattering, random events, positron range and detector dimensions limit the estimate; a diagram should mark a possible event point somewhere on the LOR.



Proposed allocation: Two detectors and LOR sketch (1); Event can lie along LOR (1); Many LORs and reconstruction needed (1); Physical limitation / count statistics (1). **Total 4.**

Source map: S06 pp.25-34. **Closest past question:** 2025 Q1(b)ii. **Status:** Recurring concept with additional reasoning.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 2: Radiography and CT beyond the common repeats

2(a) [4 marks]

Question. Explain the origin of the continuous and characteristic components of an X-ray-tube spectrum. State the different primary effects of increasing kVp and increasing mAs.

Expected answer and working. Decelerating electrons in the target nuclear field emit a continuous bremsstrahlung spectrum. Inner-shell vacancies followed by electronic transitions emit discrete characteristic lines specific to target material. Increasing kVp raises maximum photon energy and generally changes beam output/penetration; increasing mAs primarily raises photon number for a given kVp without setting a new maximum energy. Most tube electron energy becomes heat.

Proposed allocation: Bremsstrahlung (1); Characteristic transitions (1); kVp effect (1); mAs effect (1). **Total 4.**

Source map: S2 pp.7,13-15. **Closest past question:** 2025 Q2(a)i is closest tube question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [4 marks]

Question. Compare photoelectric absorption and Compton scattering for diagnostic X-rays. Explain why pair production is not significant at a maximum tube photon energy of 120 keV.

Expected answer and working. Photoelectric absorption removes the incident photon and ejects a bound electron; its strong dependence on photon energy and effective atomic number contributes to tissue/contrast-agent differences. Compton interaction ejects a loosely bound electron and redirects a lower-energy photon, adding scatter that reduces image contrast. Both can deposit energy. Pair production requires at least approximately 1.022 MeV, well above 120 keV=0.120 MeV, so it is energetically impossible for that beam.

Proposed allocation: Photoelectric mechanism (1); Contrast dependence (1); Compton mechanism and scatter effect (1); Threshold comparison (1). **Total 4.**

Source map: S2 pp.17-25. **Closest past question:** 2024 Q2(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [4 marks]

Question. Compare computed radiography (CR), indirect digital radiography and direct digital radiography by their signal-conversion and readout principles.

Expected answer and working. CR stores a latent image in a photostimulable phosphor plate and uses a laser reader to release light for digitization. Indirect DR converts incident X-rays to light in a scintillator, then to charge in a photodiode/flat-panel array. Direct DR converts X-rays to charge in a photoconductor and reads the charge electronically. Digital output does not imply that the primary conversion is direct; each approach has its own resolution and efficiency trade-offs.

Proposed allocation: CR storage and laser readout (2); Indirect X-ray→light→charge (1); Direct X-ray→charge (1). **Total 4.**

Source map: S2 pp.39-42. **Closest past question:** 2025 Q2(a)i is closest detector context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [5 marks]

Question. A mass of 0.20 kg absorbs 0.60 mJ of X-ray energy. Calculate absorbed dose in Gy. With radiation weighting factor wR=1, determine equivalent dose in Sv. If only this tissue is exposed and its tissue weighting factor is 0.12, calculate its contribution to effective dose. Distinguish these three quantities.

Expected answer and working. Absorbed dose D=energy/mass=(0.60e-3)/0.20=0.003 J/kg=0.003 Gy=3 mGy. Equivalent dose H=wR D=0.003 Sv=3 mSv for the given radiation weighting. Effective-dose contribution E=wT H=0.12·0.003=0.00036 Sv=0.36 mSv. Absorbed dose is physical energy per mass; equivalent dose weights radiation type; effective dose weights irradiated tissue contributions. Equal numerical values for X-rays do not make Gy and Sv the same physical quantity.

Proposed allocation: Energy conversion and D=3 mGy (2); H=3 mSv (1); E contribution=0.36 mSv (1); Distinguish physical and weighted quantities (1). **Total 5.**

Source map: S2 pp.54-59. **Closest past question:** No direct historical dose numerical. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(e) [4 marks]

Question. For this question define multi-detector helical pitch as table travel per rotation divided by total active collimated beam width. A scanner has 16 active rows each 1.25 mm wide, table travel 30 mm/rotation and rotation time 0.50 s. Calculate pitch and table speed; neglect end effects and find time to cover 300 mm. Explain why the pitch convention must be stated.

Expected answer and working. Total active width=16·1.25=20 mm. Pitch=30/20=1.5. Table speed=30/0.50=60 mm/s; covering 300 mm takes 300/60=5.0 s, excluding extra reconstruction/end coverage. A single-slice convention can use one collimated slice thickness; substituting 1.25 mm here would produce 24 and is not the specified multi-detector convention. The lecture gives a single-slice pitch formula, so the extension is explicitly supplied.

Proposed allocation: 20 mm width (1); Pitch 1.5 (1); 60 mm/s and 5 s (1); Convention explanation (1). **Total 4.**

Source map: S3 pp.39-41; extended pitch definition supplied in question. **Closest past question:** 2023 Q2(b)ii is closest scanner-geometry precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(f) [4 marks]

Question. Explain the causes of (i) a ring artifact in third-generation CT and (ii) cupping in a uniform phantom. Give one corrective action for each.

Expected answer and working. A persistently miscalibrated detector samples a consistent radial position as the tube/detector arc rotates, producing a ring after reconstruction; recalibration or detector service addresses it. Cupping can arise from beam hardening: lower-energy photons are preferentially absorbed and the centre attenuation is underestimated under a simple monoenergetic model. Beam-hardening calibration/correction or appropriate filtration can reduce it; scatter correction also addresses scatter-related cupping.

Proposed allocation: Ring mechanism (1); Detector calibration/service (1); Cupping mechanism (1); Appropriate correction (1). **Total 4.**

Source map: S3 pp.56-59. **Closest past question:** 2023 Q2(b)iii is closest QA context. **Status:** Original syllabus coverage.

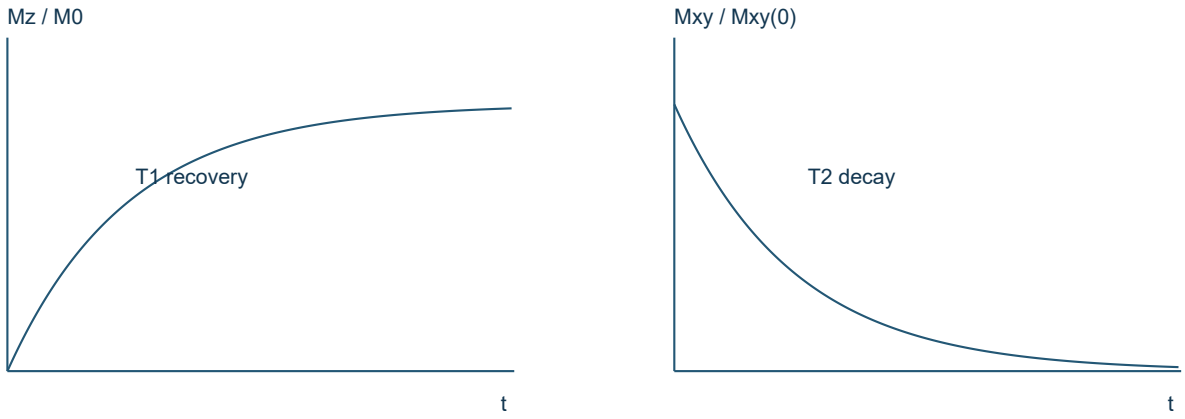
Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 3: MRI relaxation, reconstruction and functional imaging

3(a) [4 marks]

Question. After a 90° excitation, write and sketch ideal longitudinal recovery and transverse decay. Distinguish T1, T2 and T2*.

Expected answer and working. For initially saturated longitudinal magnetization Mz(0)=0, Mz(t)=M0[1-exp(-t/T1)]. The transverse envelope is Mxy(t)=Mxy(0)exp(-t/T2) when only intrinsic dephasing is considered. At t=T1, longitudinal recovery is 63.2% of M0; at t=T2, transverse signal is 36.8% of its initial value. T1 concerns spin-lattice recovery; T2 describes irreversible transverse coherence loss; T2* additionally includes reversible static-field dephasing and is no longer than T2 under the usual model.



Proposed allocation: Longitudinal equation/sketch (1); Transverse equation/sketch (1); T1 vs T2 interpretation (1); T2* distinction (1). **Total 4.**

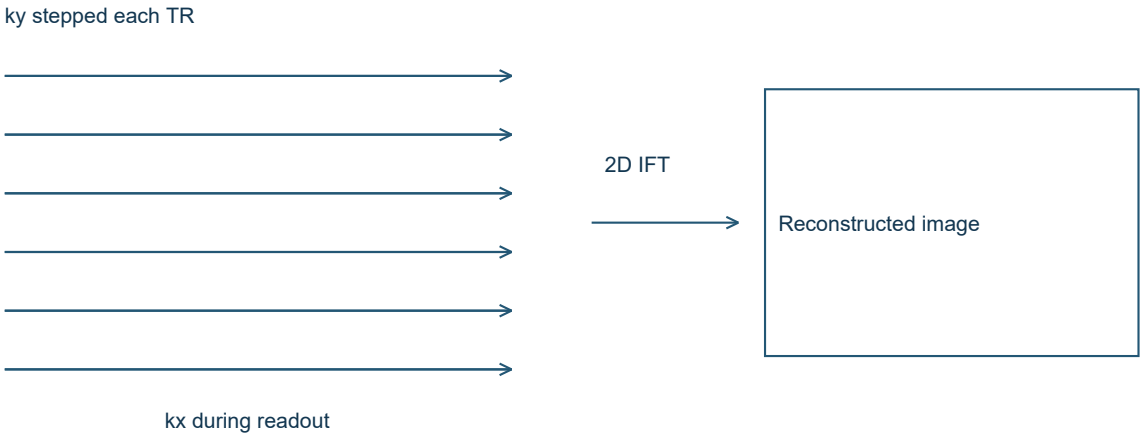
Source map: S4 pp.29-43. **Closest past question:** 2025 Q3(b)v-vi is closest relaxation mechanism. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [5 marks]

Question. Explain how phase- and frequency-encoding gradients sample a 2D k-space grid, how the image is reconstructed, and how reducing Δk or increasing the maximum sampled spatial frequency changes the image.

Expected answer and working. A phase-gradient area sets ky for a repetition; its value is stepped across repeated TRs. The read gradient makes phase accumulate continuously during ADC sampling so kx is traversed along a line. The signal samples a spatial Fourier representation of the slice; a 2D inverse Fourier transform yields the image. FOV=1/Δk, so smaller sample spacing increases field of view. Extending the sampled spatial-frequency range (larger |kmax|) supports finer detail; this is distinct from merely interpolating a grid.



Proposed allocation: Phase gradient sets ky (1); Read gradient traverses kx (1); Inverse Fourier reconstruction (1); Smaller Δk gives larger FOV (1); Larger |kmax| supports finer resolution (1). **Total 5.**

Source map: S4 pp.57-71,87-90. **Closest past question:** 2025 Q3(b)i-iii is closest encoding precedent. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [4 marks]

Question. State suitable qualitative TR and TE choices for proton-density-weighted, T1-weighted and T2-weighted conventional spin-echo images. Explain why long TR reduces T1 contrast.

Expected answer and working. Proton-density weighting uses long TR and short TE to minimize both relaxation weightings. T1 weighting uses relatively short TR and short TE so differential recovery remains. T2 weighting uses long TR and relatively long TE so transverse decay dominates. Long TR allows tissues to approach full longitudinal recovery, making (1-exp(-TR/T1)) approach 1 for both and reducing dependence on T1. Choices are relative to tissue relaxation times, not fixed universal millisecond thresholds.

Proposed allocation: PD long TR/short TE (1); T1 short TR/short TE (1); T2 long TR/long TE (1); Recovery explanation (1). **Total 4.**

Source map: S4 pp.75-81. **Closest past question:** 2024 Q3(a)i is closest intensity question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [3 marks]

Question. An object extends beyond the encoded field of view in the phase direction. Explain wraparound aliasing and give two acquisition remedies with their trade-offs.

Expected answer and working. Discrete k-space sampling makes the reconstructed image periodic with period FOV=1/Δky. Signal outside the chosen FOV folds into the displayed region. Increase phase FOV by reducing Δky, with enough additional phase steps to preserve resolution (longer scan), or use phase oversampling with later cropping (more data/time). Spatially selective excitation/saturation is another acceptable remedy when applicable. Merely adding interpolated display pixels does not remove true aliasing.

Proposed allocation: Periodic sampling / folding explanation (1); Valid remedy with trade-off (1); Second valid remedy with trade-off (1). **Total 3.**

Source map: S4 pp.87-91. **Closest past question:** No direct historical aliasing question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [5 marks]

Question. Explain the BOLD basis of fMRI, why rapid EPI acquisition is useful, and how a simple repeated task/rest experiment identifies associated regions. State one interpretation limitation.

Expected answer and working. Neuronal activation changes local haemodynamics; increased oxygenated blood flow can reduce local deoxyhaemoglobin concentration. Deoxyhaemoglobin is paramagnetic, so the changed susceptibility alters transverse dephasing and commonly increases T2*-weighted GE signal in activated regions. EPI provides rapid repeated whole-brain sampling. Record repeated task/rest blocks and test voxel signal association with the expected haemodynamic task response, accounting for delay rather than assuming instant neural timing. BOLD is an indirect vascular proxy and is affected by motion and physiology; correlation is not direct measurement of neural firing.

Proposed allocation: Activation and oxygenation/deoxyhaemoglobin (1); Susceptibility and MR signal mechanism (1); Rapid EPI rationale (1); Repeated blocks / voxel association (1); Indirect/delayed or motion limitation (1). **Total 5.**

Source map: S4 pp.84-85,105-108; syllabus p.1. **Closest past question:** No direct historical fMRI question. **Status:** Original explicit-syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [4 marks]

Question. Explain time-of-flight MR angiography in terms of inflowing blood and saturated stationary tissue. Why can a large tip angle relative to the stationary tissue Ernst angle improve vessel contrast, and what is one limitation?

Expected answer and working. Rapid repeated GE excitations partially saturate stationary spins. Fresh blood entering the slab has experienced fewer RF pulses and has more longitudinal magnetization, giving a stronger signal than stationary background. A relatively large angle further suppresses the stationary steady-state signal while fresh spins can still respond strongly. Slow flow, in-plane flow or repeated excitation through a thick slab can saturate blood too, reducing vessel brightness. The explanation concerns inflow history, not a universal intrinsic shortening of blood T1.

Proposed allocation: Stationary saturation (1); Fresh inflow signal (1); Relative tip-angle contrast reasoning (1); Slow/in-plane flow limitation (1). **Total 4.**

Source map: S4 pp.102-104. **Closest past question:** No direct historical MR-angiography question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 4: Ultrasound instrumentation, OCT and thermography

4(a) [4 marks]

Question. Explain piezoelectric transmission and reception in an ultrasound probe. Sketch a basic pulse-echo system signal path including a transmitter/receiver switch and display.

Expected answer and working. An applied alternating electric field causes the piezoelectric element to expand and contract, launching sound (inverse effect). Returning pressure deforms the element and produces an electrical response (direct effect). A transmit/receive switch protects and routes the receiver during high-voltage firing. Receive amplification, depth-dependent gain, envelope detection and scan conversion produce a brightness image. Coupling, timing and acoustic backing/focusing influence the response.



Receive path returns through T/R switch; schematic shows functional sequence

Proposed allocation: Inverse effect / transmission (1); Direct effect / reception (1); T/R switch and receive processing (1); Labeled signal path (1). **Total 4.**

Source map: S05 pp.17-21,40. **Closest past question:** 2024 Q4(a)iv is closest system context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [4 marks]

Question. Compare linear and phased arrays for beam formation. Explain how electronic focusing improves lateral resolution and give one cost of using several transmit focal zones.

Expected answer and working. A linear array typically uses successive subapertures to produce approximately parallel lines and a rectangular image. A phased array applies inter-element delays across an aperture to steer a sector beam. Curved delay profiles align wave arrival at a chosen focus, narrowing the beam and improving lateral resolution there; receive delays can vary with echo depth. Several transmit focal zones require additional firings and reduce frame rate at fixed depth and line count.

Proposed allocation: Linear subaperture scan (1); Phased steering delays (1); Focusing/narrower beam (1); Multiple-focus frame-rate cost (1). **Total 4.**

Source map: S05 pp.23-31. **Closest past question:** 2024 Q4(a)iv; 2025 Q4(a)iii. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [3 marks]

Question. Explain what Doppler ultrasound adds to B-mode imaging. State how motion towards versus away from the probe affects the detected frequency, and why perpendicular flow is difficult to measure with this principle.

Expected answer and working. B-mode displays structure from echo strength; Doppler measures frequency shifts associated with moving scatterers, commonly blood, and provides flow-related functional information. Motion towards the probe raises the received frequency and motion away lowers it under the usual backscatter convention. The shift depends on the velocity component along the beam; perpendicular motion has approximately zero component and little Doppler shift.

Proposed allocation: Flow information beyond structure (1); Towards/away shift directions (1); Beam-axis component / perpendicular limitation (1). **Total 3.**

Source map: S05 pp.3,39; syllabus p.1. **Closest past question:** No direct historical Doppler question. **Status:** Original explicit-syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [2 marks]

Question. State two physical reasons why ultrasound may fail to provide a useful image of a target behind an air-filled cavity or thick bone.

Expected answer and working. The large acoustic-impedance mismatch causes strong reflection and poor transmission through the interface. Bone and air also strongly attenuate the propagating beam, so target echoes can fall below the detector range and posterior acoustic shadowing results. Changing gain cannot reconstruct echoes that never reach the receiver.

Proposed allocation: Impedance/reflection barrier (1); Attenuation/weak deep echoes (1). **Total 2.**

Source map: S05 pp.4,13-16,38. **Closest past question:** 2025 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [6 marks]

Question. Compare time-domain and Fourier-domain OCT. Explain A-scan-to-B-scan formation, give one ophthalmic application, and identify one acquisition or processing error that could corrupt a thickness measurement.

Expected answer and working. Time-domain OCT scans reference delay mechanically to sample depth sequentially. Fourier-domain OCT obtains depth information from an interference spectrum: a spectrometer is used in spectral-domain systems, or a swept source plus detector in swept-source systems; Fourier processing recovers the depth profile. Lateral juxtaposition of A-scans forms a B-scan. Retinal-layer/nerve-fibre-layer monitoring is an appropriate application. Eye motion, poor focus, blink dropout or incorrect automated layer segmentation can corrupt measured thickness, so image and segmentation boundaries require review.

Proposed allocation: Time-domain reference-delay scan (1); Fourier-domain spectral acquisition (1); Fourier recovery of depth (1); A/B scan formation (1); Ophthalmic application (1); Error + effect on measurement (1). **Total 6.**

Source map: A2 PDF pp.3-8; syllabus p.1. **Closest past question:** 2024 Q4(b)ii; 2025 Q4(b)iii. **Status:** Historical basics plus assignment-supported coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [6 marks]

Question. A thermal camera compares a body surface at 307 K with a reference surface at 305 K. Assuming equal emissivity and the supplied total-emission law $M=\varepsilon\sigma T^4$, calculate the percentage increase of emitted power. Explain why the change is not by itself a diagnosis, and give three controls for a reliable comparison.

Expected answer and working. Equal ε and σ cancel: $M_{307}/M_{305}=(307/305)^4=1.02648863$, so emitted power rises by about 2.64886%. This calculation describes total emitted radiant power, not a calibrated band-limited camera voltage directly. Temperature patterns are nonspecific and affected by environment and perfusion; a change alone cannot establish a particular disease. Controls include a calibrated reference/check of detector uniformity, stable room conditions with patient acclimatization, and consistent emissivity/view angle/distance/region positioning. Use thermal findings with clinical context.

Proposed allocation: T^4 ratio setup (1); 2.64886% result (1); Nonspecific / band-limited interpretation (1); Calibration control (1); Environmental/patient control (1); Geometry/emissivity control (1). **Total 6.**

Source map: A1 PDF pp.4-5,8-9; syllabus p.1. **Closest past question:** 2023 Q4(b)i is closest principle precedent. **Status:** Original assignment-supported numerical.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.